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MILITARY SPECIFICATION

DUCT AND DUCT ASSEMBLY, PNEUMATIC, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers a flexible pneumatic duct and duct assembly for use in the delivery of high-flow, low-pressure air at temperatures to 500°F.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

TT-I-735	Isopropyl Alcohol
TT-S-735	Standard Test Fluids, Hydrocarbon
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed And Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft And Ordnance

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-C-8188	Corrosion-Preventive Oil, Gas Turbine Engine, Aircraft Synthetic Base
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking OF U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-190	Identification Marking Of Rubber Products
MIL-STD-810	Environmental Test Methods
MS17833	Fitting, Pneumatic Starter Duct-Flanged (Male)
MS17835	Fitting, Pneumatic Starter Duct-Flanged (Female)
MS21920	Clamp, Hose (Flat Band, 500°F)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials

ASTM-D413	Methods of Test for Adhesion of Vulcanized Rubber (Friction Test)
ASTM-D629	Methods for Quantitative Analysis of Textiles (Tentative)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

3. REQUIREMENTS

* 3.1 Qualification. The duct or duct assembly furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on, or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.3 Materials. Materials and parts not specifically covered herein by specifications, standards and Government certification and approval of processes and equipment but which are necessary for the execution of this specification. shall be sound, of uniform quality and condition, and free of defects that may adversely affect the appearance, strength, endurance, or wear resistance of the finished part. The materials shall not be hammered, filed, or otherwise treated to conceal defects therein.

* 3.3.1 The materials used in the duct and duct assembly shall be resistant to deterioration or corrosion, liquid absorption or penetration, fungus growth, and sunshine to the extent specified herein.

* 3.3.2 Toxicity. The duct and duct assembly shall be constructed of materials that will not emit toxic or unpleasant odors, smoke, or fumes.

* 3.4 Design and construction. The duct assembly shall consist of a length of duct as specified by the procuring activity (see 6.2), one male and one female flange as specified in 3.4.5, and the fittings necessary to assemble the complete duct. The duct shall consist of a single integral silicone inner tube bonded to one or more layers of fabric reinforcement. A removable scuff-resistant jacket may be used if required to meet the performance requirements specified herein. If multiple lengths are used to make up the desired overall length, they shall be interchangeable. Unless otherwise specified herein, the individual duct lengths shall be optional consistent with long life and economy.

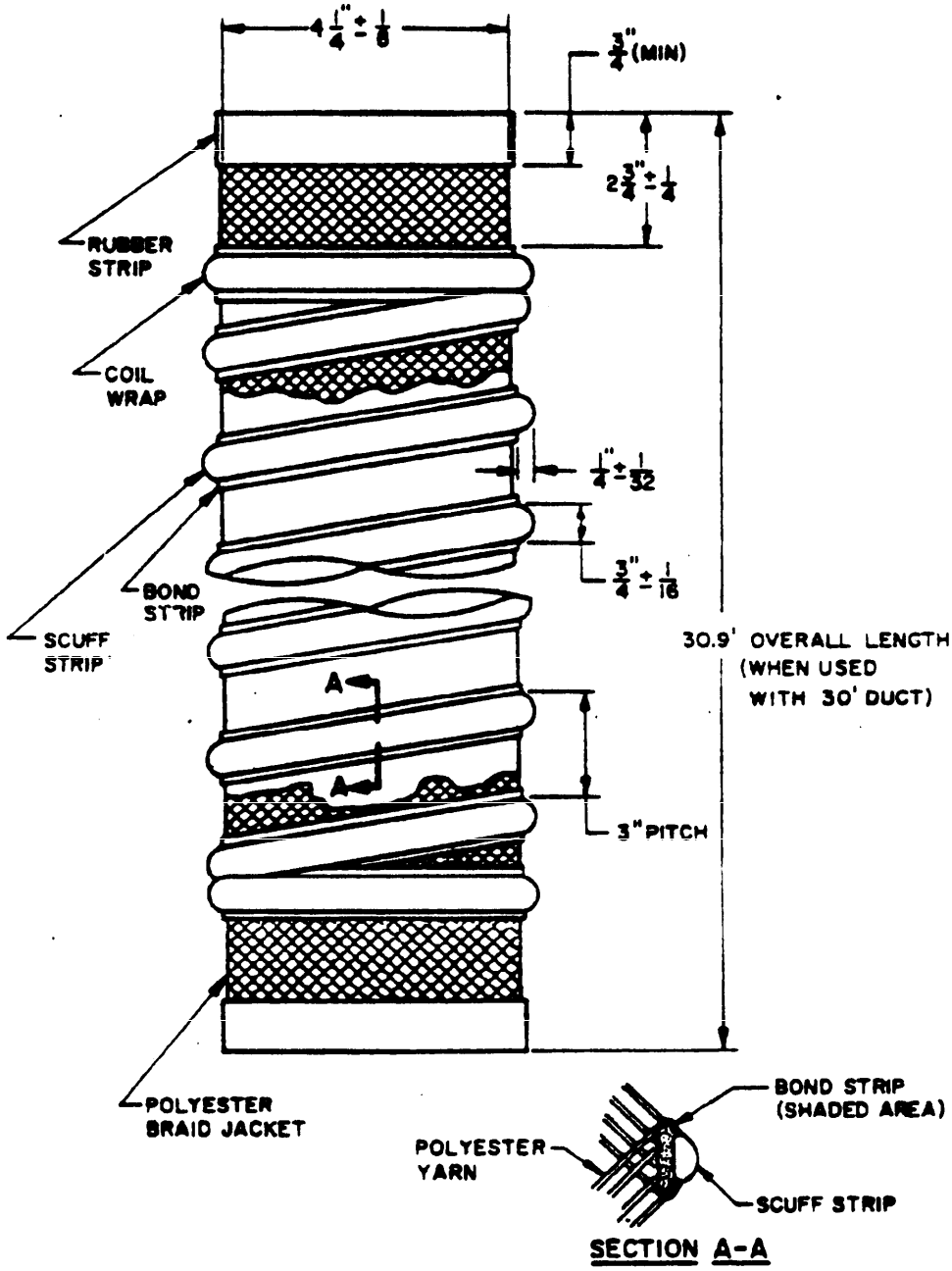
* 3.4.1 Duct diameter. The inside diameter of the duct shall be not less than 3-1/2 inches nor greater than 3-5/8 inches under standard atmospheric conditions. The outside diameter shall be as small as possible consistent with the performance requirements specified herein but shall not exceed 5 inches including the removable scuff resistant jacket. The duct construction shall be uniform throughout so that it will be possible to cut off a section of the duct at any location and reconnect to any of the standard beaded shank end fittings. No painting or other special treatment of the cut surfaces shall be required.

* 3.4.2 Serviceability. The duct and duct assembly shall be designed and constructed to resist abrasion, exposure to fluids, and leakage, and shall withstand continuous flexing consistent with the performance requirements specified herein.

3.4.3 Dimensional changes. The duct shall be so designed and constructed that it will not decrease in length more than 2 percent nor increase in length more than 3 percent when new or after exposure to any of the conditions specified herein. The inside diameter shall not decrease more than 2 percent after exposure to any of the conditions specified herein. Increases in outside diameter shall be restricted to the maximum diameter specified in 3.4.1.

* 3.4.4 Scuff resistant jacket. The scuff-resistant jacket (Figure 1) shall be constructed of a moisture resistant fabric having a maximum moisture regain of 0.5 percent as measured by ASTM Test No. D629.59T. Materials may be treated with a moisture resistant compound to meet this requirement, if necessary. The outermost jacket shall not be a material which will soil clothing. The jacket fabric shall be braided in a plain loose weave designed to form a continuous seamless cylinder. The yarn shall be treated with a bonding agent to promote required adhesion.

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SCUFF RESISTANT JACKET
FIGURE 1

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* 3.4.4.1 Scuff strip. The jacket shall be wound with an abrasion resistant protective helical coil of natural or synthetic rubber, permanently vulcanized to the fabric. A bonding strip shall be used between scuff strip and the fabric. The bond strip shall be vulcanized to the jacket so that the jacket fabric immediately below the bond strip is completely contained within the rubber material and there are no exposed threads below the bond strip. The last coil wrap at each end shall be wrapped squarely around the jacket and vulcanized to itself to prevent separation or raveling of the strip.

* 3.4.4.2 Jacket ends. Each end of the jacket fabric shall be vulcanized with 3/4 inch minimum width rubber strip to prevent raveling of the jacket material.

* 3.4.5 End flanges. Each complete duct assembly shall have one male and one female fitting conforming to MS17833 and MS17835, respectively, attached with two clamps at each end conforming to MS21920.

3.4.6 Fittings. All end fittings, sleeves, splices, and attaching devices shall be so designed that they will not require adjustment nor replacement after exposure to any of the conditions specified herein.

* 3.4.7 Wall thickness. The duct shall have a uniform wall thickness with a uniform tolerance of ±1/32 inch.

* 3.5 Performance. The duct assembly shall be capable of satisfying the performance requirements of Section 4 and of withstanding the following conditions:

a. An ambient temperature of -65° Fahrenheit (F) when flowing air at 250 pounds per minute, 90 pounds per square inch absolute (psia), and 300°F.

b. An ambient temperature of 130°F when flowing air at 100 pounds per minute, 65 psia, and 500°F.

c. A static proof pressure of 250 pounds per square inch (psi) at ambient temperatures.

d. Exposure, but not operation, at -80°F ambient temperature.

e. Exposure to salt-sea atmosphere.

f. Fungus growth as encountered in tropical climates.

g. Radiant energy as found under natural conditions.

h. A burst pressure of 400 psi at ambient conditions.

i. A minimum bend radius of the duct of 8 inches under no pressure and 20 inches while under 30 psi.

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* 3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

* 3.7 Weight. The weight of the duct without end flanges and fittings but with scuff cover shall not exceed 2.5 pounds per foot of duct length.

3.8 Marking.

* 3.8.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. In addition, the scuff resistant jacket shall be marked with manufacturer's name, part number, and cure date to be either embossed or marked in white in accordance with MIL-STD-190.

* 3.8.2 Date of manufacture. The date of manufacture shall be permanently marked in a visible location on each end of the duct. The marking shall be the quarter of the year and the year of manufacture.

* 3.9 Workmanship. The duct and duct assembly shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom of blemishes and defects and marking of parts.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the duct and duct assembly shall be classified as follows:

a. Qualification tests . . . . See 4.3

b. Acceptance testing . . . . See 4.4

* 4.3 Qualification testing. Qualification testing shall consist of all tests listed below:

Test	Paragraph (Requirement Paragraph)
a. Examination of Product	4.5.1. (3.4)
b. Leakage	4.5.2 (3.4.2)
c. Static Proof Pressure	4.5.3 (3.5c)
d. Low Temperature	4.5.4 (3.4.2, 3.5a and d)
e. Bend Radius and Length	4.5.5 (3.4.3 and 3.5i)

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f.	High Temperature	4.5.6 (3.5b)
g.	Temperature Drop	4.5.7 (3.5b)
h.	Pressure Drop	4.5.7 (3.5b)
i.	Burst Pressure	4.5.9 (3.5h)
j.	Abrasion Resistance	4.5.10 (3.4.2)
k.	Adhesion	4.5.11 (3.4)
l.	Endurance	4.5.12 (3.4.2 and 3.5b)
m.	Fluids Resistance	4.5.13 (3.4.2)
n.	Environmental	4.5.14 (3.5e, f and g)
o.	Scuff Resistant Jacket	4.5.15 (3.4.4) (if applicable)

* 4.3.1 Qualification test samples. The qualification test samples shall consist of three duct assemblies 30 foot in length. If qualification approval is desired for the duct only, end flanges and fittings necessary to complete the duct assembly shall be furnished by the manufacturer. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

* 4.3.2 Test sequence. The samples specified in 4.4.1 herein for qualification testing shall be submitted to those tests as follows, with the tests conducted in the order listed:

SAMPLE 1

Examination of product	4.5.1
Leakage	4.5.2
Static proof pressure	4.5.3
Low temperature	4.5.4
Bend radius and length	4.5.5
High temperature	4.5.6
Burst pressure	4.5.9

SAMPLE 2

Examination of product	4.5.1
Leakage	4.5.2
Static proof pressure	4.5.3
Adhesion	4.5.11
Solvent resistance	4.5.13
Environmental	4.5.14
Scuff jacket (if used)	4.5.15

SAMPLE 3

Examination of product	4.5.1
Leakage	4.5.2
Static proof pressure	4.5.3
Temperature drop	4.5.7
Pressure drop	4.5.10
Endurance	4.5.12

* 4.3.3 Test report. When the tests are conducted as specified in Section 4, a test report in accordance with MIL-STD-831 shall be furnished to the specification preparing activity (see 6.3).

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* 4.3.4 Test methods. The qualification tests shall follow the methods prescribed in 4.5. The qualification tests may, at the option of the preparing activity, be supplemented by tests under actual service conditions. If a scuff-resistant jacket is used, it shall be considered an integral part of the duct and all tests conducted on the duct under this specification shall be conducted on the entire assembly.

* 4.3.5 Qualification required. Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

* 4.4 Acceptance tests. The acceptance tests shall consist of the following tests:

a. Individual tests . . . . See 4.3.1

b. Sampling tests . . . . . See 4.3.2

* 4.4.1 Individual tests. Each duct shall be subjected to the following tests as described under 4.5:

a. Examination of product . . See 4.5.1

b. Leakage . . . . . See 4.5.2

* 4.4.2 Sampling tests. One sample duct assembly selected at random from each lot of 150 ducts or fraction thereof shall be subjected to the following tests:

Adhesion 4.5.11

Dynamic flexing 4.5.12.2

In addition, if a scuff resistant jacket is used, the sample shall undergo the tests of 4.5.15.

* 4.5 Test methods.

* 4.5.1 Examination of product. The duct or duct assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, weight, and dimensions.

* 4.5.1.1 Upon completion of each test specified herein, the duct shall be disassembled and inspected. Any cracking, ballooning, or material separation shall be cause for rejection.

* 4.5.2 Leakage. Allowable leakage rates shall be specified in terms of a 30-foot duct. The test shall be conducted at ambient prevailing temperatures. With the test duct blocked at both ends, it shall be pressurized with air to 150 psi, and, with the pressure trapped, it shall not drop to 100 psi in less than 1 hour.

* 4.5.3 Static proof pressure. The duct assembly shall be sealed at one end and subjected to a static water pressure of 250 psi at normal prevailing ambient temperatures to insure compliance with the static proof pressure requirement specified under 3.5. There shall be no indication of failure or impending failure as a result of this test.

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• 4.5.4 Low temperature. The duct assembly shall be placed in a chamber and cooled to, and maintained at, a temperature of -80°F for a period of 24 hours. Permanent damage to the duct as a result of this test shall be cause for rejection. The temperature shall be raised to, and maintained at, -65°F for a period of 24 hours. While at this temperature, the duct shall be flexible. At the end of the 24-hour period and with the temperature maintained a 2,000-pound, rubber-covered weight shall be placed on the duct. The weight shall cover 6 inches of duct length and overhang the width of the duct when flattened. The weight shall be removed after 1 minute. This test shall be repeated 25 times examining the duct each time the weight is removed. Permanent damage to the duct shall be cause for rejection.

• 4.5.4.1 Flexing. Immediately upon completion of the low temperature test, and with the temperature maintained at -65°F, the duct assembly shall be flexed by reeling spirally on a drum. The drum shall be a maximum of 12 inches in diameter and long enough to allow a 30 foot duct to be coiled on the drum without overlapping of the windings. The first flexing shall be accomplished by clockwise rotation of the drum. The duct assembly shall then be unreeled to its original position after which it shall be reeled back on the drum by counterclockwise rotation of the drum. This shall constitute one flexing cycle. The duct assembly shall be subjected to one hundred flexing cycles after which it shall be subjected to the test specified in 4.5.2 and 4.5.3. Upon satisfactory completion of the flexing cycle, the duct shall be examined internally and at all attaching adapters for breaks and cracks. Any change shall be photographed and any unusual occurrences recorded. With the temperature maintained at -65°F, air at 300 ±5°F, 90 ±2 psia and 250 ±5 pounds per minute shall be supplied to the duct for a period of 5 minutes. The duct shall then be allowed to cool for a period of fifty-five minutes. This shall constitute one cycle. The duct assembly shall be subjected to a total of 25 such cycles. Upon completion of the test, the duct assembly shall be subjected to the tests specified in 4.5.2 and 4.5.3.

• 4.5.5 Bend radius and length. At a stabilized -65°F temperature, the unpressurized duct assembly shall be subjected to bending until it will not bend farther without being damaged. The minimum bend radius shall not exceed 8 inches as measured to the centerline of the duct. Subsequently, the duct assembly shall be pressurized with air to 30 psia and the bend radius redetermined. The bend radius shall not exceed, without crimping, 20 inches as measured to the centerline of the duct. The temperature shall then be stabilized at 130°F and the test (unpressurized and pressurized) repeated. After the pressurized bend radius has been determined, the pressure shall be decreased to 15 psia in increments of 5 psia and the change in the duct length determined. The change in duct length shall not be greater than specified in 3.4.3. The duct assembly shall then be subjected to the tests specified in 4.5.2 and 4.5.3.

• 4.5.6 High temperature. The duct assembly shall be placed within a chamber and the ambient temperature maintained at 130 ±2°F for a period of 24 hours. At the end of this time, the duct shall be subjected to a static load of 2,000 pounds as specified in 4.5.4 herein repeated 25 times.

• 4.5.7 Temperature drop. A 30 foot duct assembly shall be subject to air supplied at 450 ±5°F 70 ±1 psia and 120 ±5 pounds per minute. The ambient temperature for the test shall be 75 ±5°F. Temperatures at the center of the airstream shall be measured at each end of the duct. The temperature drop through a 30 foot duct assembly shall not exceed 30°F.

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4.5.8 Pressure drop. A 30 foot duct assembly shall be subjected to the flow conditions of 4.5.7. The pressure drop shall not exceed 6 psi across the length of the duct assembly.

4.5.9 Burst pressure. The duct assembly shall be sealed at one end and water pressure applied in 10-pound increments until the duct assembly bursts. This test shall be accomplished at normal prevailing ambient temperatures. Any irregularities in cross section or any localized ballooning shall be noted. Any failures shall be photographed. The minimum acceptable burst pressure shall be 400 psia.

4.5.10 Abrasion resistance. The 30 foot sample shall be bent 180 degrees so that the point of bend occurs at approximately one-eighth of the total length. The duct assembly shall be attached to a towing vehicle at a point approximately 6 inches above the concrete, at approximately 40 inches from the point of bend. The point of bend shall rest upon the concrete, (a dry, brushed surface subject to approval by the procuring activity). The duct assembly, thus supported, shall be dragged at a minimum of 5 MPH for 20,000 feet without rotation. This test shall be accomplished three times with the point of contact being changed 3-3/4 feet for each test. Following each drag test, the duct assembly shall be tested in accordance with 4.5.2 and 4.5.3. In addition, there shall be no signs of localized ballooning as a result of this test.

4.5.11 Adhesion test. The test and test apparatus shall be of a dead weight method as designated in ASTM specification D413 with the following exceptions. The test specimen strip shall be 1-1/2 inches wide. The adhesion of the fabric reinforcement to the tube shall not be less than 10 pounds. The rate of separation shall not exceed 1 inch in one minute.

4.5.12 Endurance test.

4.5.12.1 Flow. The duct assembly shall be cycled at the 130°F ambient temperature with air flowing at 100 ±10 pounds per minute, 65 ±2 psia total pressure and 500 ±5°F inlet temperature. A cycle shall consist of 4 hours under flow conditions and 1/2 hour under no-flow conditions. During this depressurized time, the duct assembly shall be examined for deterioration. Any failure or impending failure shall be cause for rejection. This procedure shall be continued until 25 cycles have been accumulated. The flanges shall not be adjusted at any time during this test. The duct assembly shall then be subjected to 500 dynamic flexing cycles in accordance with 4.5.12.2. Upon completion of these tests, the duct assembly shall be tested in accordance with 4.5.2 and 4.5.3.

4.5.12.2 Dynamic flexing. A 10-foot section of duct assembly shall be mounted in a test rig and flexed 1,000 times. The test rig shall be constructed to hold the 10-foot section of duct assembly in a vertical position while under airflow conditions. The rig shall be of sufficient height to allow a 10-foot section of duct assembly placed in the flexing position to hang with one end touching the ground. Not more than 6 inches of this end shall be in contact with the ground. Air shall be supplied to the duct for 30 seconds followed by a no-flow condition of 20 seconds. At the start of the measured 30-second flow condition, the air temperature flowing into the inlet of the test duct shall be 475 ±5°F, the pressure shall be 70 ±2 psia, and the flow shall be 75 ±5 pounds per minute. The pressure conditions shall be attained in from 3 to 7 seconds and air supply cutoff shall occur in 1/2 second or less. When air is supplied to the duct, the free end of the duct assembly shall rise unrestrained to a

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height of 7 feet. At this point, the free end shall be restrained (by rope or other suitable means) to prevent whipping of the duct assembly and to insure that it falls to the original position when the air supply is cut off. Following the flexing cycles, the duct assembly shall be tested in accordance with 4.5.2 and 4.5.3.

* 4.5.13 Fluid resistance. Adjacent one-foot lengths of the duct exterior shall be saturated with each of the following fluids for a 5-minute period:

    a. Test fluid conforming to TT-S-735, Type III.

    b. Test fluid conforming to TT-S-735, Type IV

    c. Engine lubricating oil conforming to MIL-L-7808.

    d. Grade B lubricating oil conforming to MIL-C-8188.

    e. Anti-icing fluid conforming to TT-I-735.

    f. Hydraulic fluid conforming to MIL-H-5606.

  This saturation shall be repeated 5 times at approximately 24 hour intervals, after which it shall be tested in accordance with 4.5.2 and 4.5.3.

* 4.5.14 Environmental. The following environmental tests shall be conducted in accordance with the specified procedures of MIL-STD-810.

* 4.5.14.1 Fungus resistance. The fungus resistance test shall be conducted in accordance with Method 508.

* 4.5.14.2 Salt spray. The salt spray test shall be in accordance with Method 509 (couplings only).

* 4.5.14.3 Sunshine. The sunshine test shall be in accordance with Method 505, Procedure I for a period of 100 hours.

* 4.5.15 Scuff resistant jacket. If a scuff resistant jacket is used as part of the duct and duct assembly, it shall be subjected to the tests of 4.5.15.1 and 4.5.15.2

* 4.5.15.1 Pull test. The scuff resistant jacket shall be capable of withstanding a 1500 pound pull applied axially at the ends of the jacket. Plugs shall be installed and clamped into each end of the jacket. The plugs shall have attached hooks and one end shall be hooked to a ceiling support while a 1500 weight is suspended from the other end for a minimum of 5 minutes. Any separation of components, breaks in the braid jacket, or breaks in the scuff strip shall be cause for rejection. This test shall be repeated after heating the scuff resistant jacket to 200°F for 24 hours.

* 4.5.15.2 Scuff strip adhesion test. The scuff strip shall be incapable of being separated from the bond strip along an apparent parting surface. Separation shall be initiated by slicing with a knife at the scuff strip - bond strip interface.

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The two strips shall then be pulled away from each other. If separation continues along an apparent parting surface, this shall be cause for rejection. Any separation of the scuff strip - bond strip shall occur as tearing of the rubber rather than by pulling apart of one distinct strip from the other.

* 4.6 Inspection of the preservation, packaging, and packing. The inspection of the preservation, packaging, and packing shall be in accordance with the instructions in Section 5, or the specification referenced therein.

5. PACKAGING

5.1 Preservation and packaging.

* 5.1.1 Level A. Ducts and duct assemblies shall be preserved in accordance with MIL-P-116, method III, using a snug-fitting, class weather-resistant container conforming to PPP-B-636.

5.1.2 Level C. Ducts and duct assemblies shall be preserved and packaged in such manner as to insure adequate protection during shipment to the initial destination for immediate use.

5.2 Packing.

* 5.2.1 Level A. Ducts and duct assemblies shall be packed in snug-fitting shipping containers conforming to PPP-B-576, PPP-B-585, PPP-B-601, or PPP-B-621. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with applicable container specification or appendix thereto.

* 5.2.2 Level B. Ducts and duct assemblies shall be packed in domestic-type exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations or other common carriers, as applicable to the mode of transportation.

* 5.2.4 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186 except that waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.3 Marking for shipment Interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

* 6.1 Intended use. The duct and duct assembly covered by this specification is intended for use in the delivering of high-flow, low-pressure air to aircraft starting, hydraulic and electrical power, and air conditioning systems.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Length of duct (see 3.4)

c. Level of preservation and packaging, and packing (see Section 5).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is AFLC CASO/LODS, Federal Center, Battle Creek MI 49016, and information pertaining to qualification of products may be obtained from that activity.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Review Activity:
DCSC-CS

Preparing Activity:
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